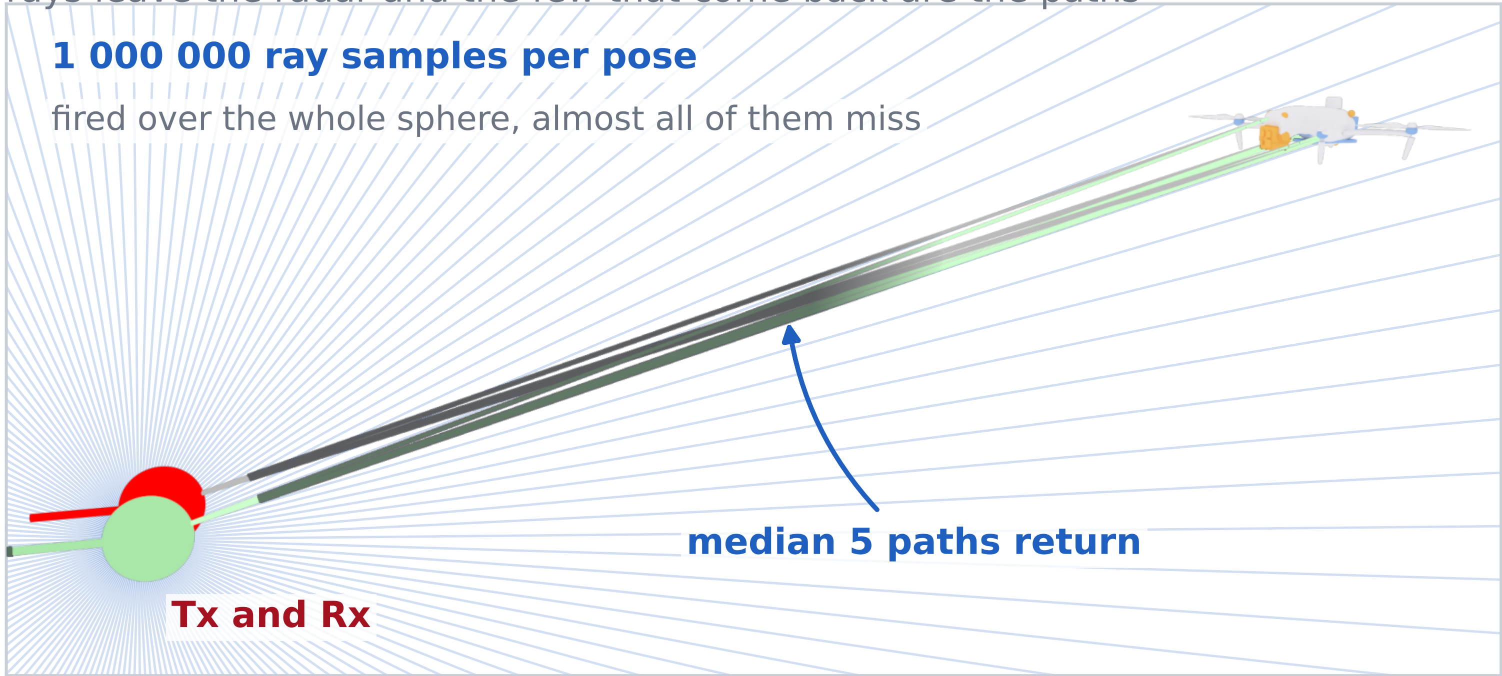
What the two engines actually compute

Both give one complex number per rotor pose. The map is what the same STFT makes of those numbers.

Sionna PathSolver

rays leave the radar and the few that come back are the paths

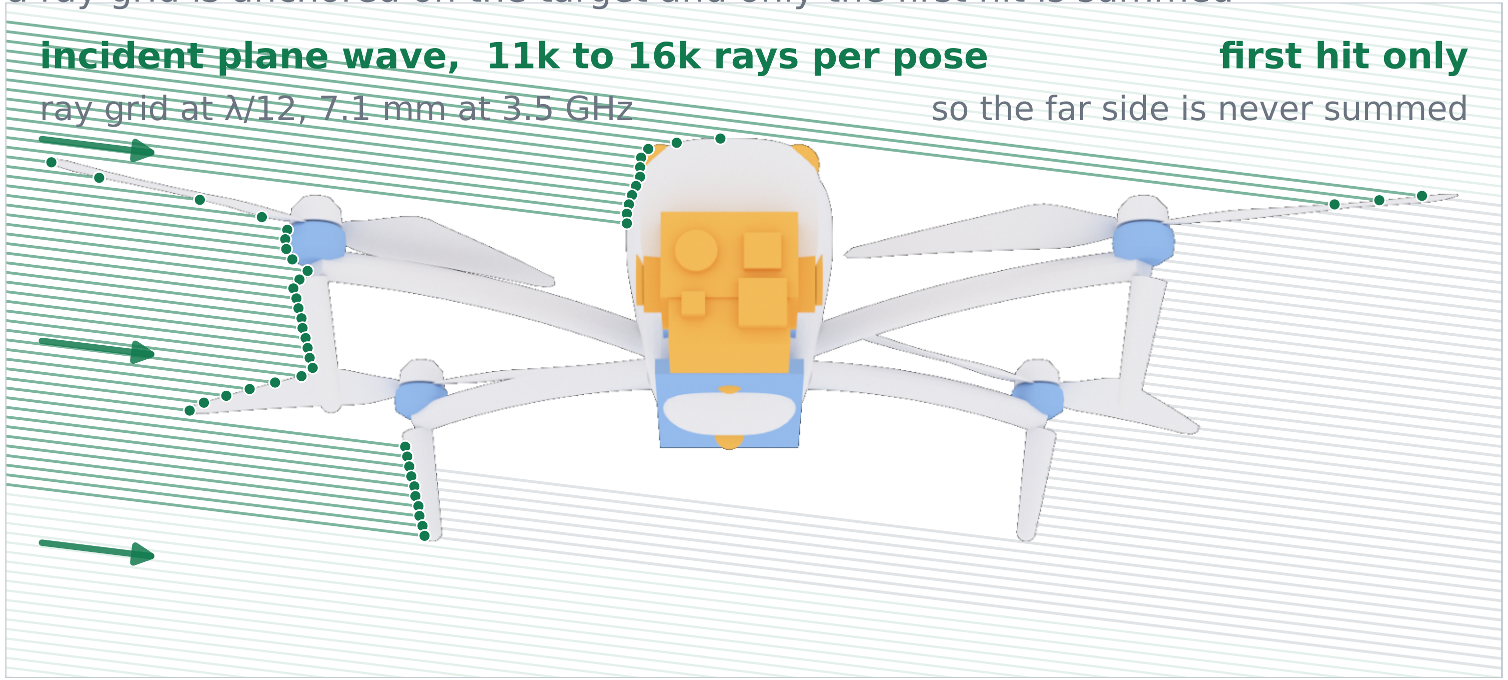


FIRED	1 000 000 ray samples, uniform on the sphere
KEPT	the paths that return, median 5 at 3 m
LEFT OUT	the surface integral over the target

Solid angle falls as $(D/R)^2$, so the ray budget grows with range

Ours (SBR+PO)

a ray grid is anchored on the target and only the first hit is summed



FIRED	a plane wave on a λ/12 ray grid
KEPT	the first hit of every ray
LEFT OUT	back facing and hidden facets

The grid rides on the target, so range is not a variable at all

BOTH COLUMNS THEN RUN THE SAME SLOW TIME CHAIN

one complex number per rotor pose → 4096 poses on a slow time grid, PRF 19.7 kHz, 208 ms → STFT, 70 sample Hann, 0.45 blade periods, hop 2 → the micro-Doppler map, time against Doppler

Real Sionna renders of the simulated DJI Matrice 4E. Rays are drawn sparse for readability and the first hits on the right are traced against the rendered silhouette.